

Atividade de Servidor DHCP

**Documentação passo a passo para configurar
e instalar um servidor DHCP:**

1º Passo

*Colocando um **IP** fixo na placa de rede secundária:*

sudo nano /etc/netplan/50-cloud-init.yaml

*Após digitar na linha de comando, sua senha será requisitada,
assim que digitar e a senha estiver correta você irá acessar a tela
de configuração da placa de rede (**imagem abaixo**):*

```
GNU nano 7.2 /etc/netplan/50-cloud-init.yaml
network:
  version: 2
  ethernet:
    enp0s3:
      dhcp4: true

aluno@aluno:~$ s
```

2º Passo

Configurar a placa de rede (ignore o erro de escrita):

```
enp0s3:
  dhcp4: true
enp0s8:
  dhcp4: no
  addresses: [172.16.0.0/24]

aluno@aluno:/$ sudo netplan apply
usage: /usr/sbin/netplan [-h] [--debug] ...
/usr/sbin/netplan : error: argument : invalid choice: 'apply' (choose from 'help', 'apply', 'generate', 'get', 'info', 'ip', 'set', 'rebind', 'status', 'try')
aluno@aluno:/$ sudo netplan apply
aluno@aluno:/$
```

3º Passo

(Observação: Sempre siga a indentação: “enp0s8” sempre na mesma linha do “enp0s3” e “dhcp4” sempre na mesma linha do primeiro “dhcp4”).

Não faça de qualquer jeito, siga exatamente o exemplo da imagem e irá funcionar)

*Depois de configurar a placa de rede, salve (**Ctrl + O**), dê um **Enter** para terminar de salvar no arquivo e depois **Ctrl + X** para sair do arquivo.*

*Após a configuração no **netplan**, aplique as alterações na máquina usando: **sudo netplan apply***

4º Passo

*Agora, instale o servidor DHCP: **sudo apt install isc-dhcp-server***

*Quando instalar, faça uma cópia do arquivo de configurações do servidor DHCP, chamado “**dhcp.conf.bkp**”:*

sudo cp /etc/dhcp/dhcpd.conf /etc/dhcp/dhcpd.conf.bkp

E entre no diretório original de configuração do DHCP:

sudo nano /etc/dhcp/dhcpd.conf

5º Passo

Quando a tela de configurações abrir, verá várias linhas comentadas com “#”.

Procure uma linha que esteja escrito:

A slightly different configuration for an internal subnet.

E então configure:

```
GNU nano 7.2 /etc/dhcp/dhcpd.conf

# No service will be given on this subnet, but declaring it helps the
# DHCP server to understand the network topology.
#
#subnet 10.152.187.0 netmask 255.255.255.0 {
#}

# This is a very basic subnet declaration.
#
#subnet 10.254.239.0 netmask 255.255.255.224 {
# range 10.254.239.10 10.254.239.20;
# option routers rtr-239-0-1.example.org, rtr-239-0-2.example.org;
#}

# This declaration allows BOOTP clients to get dynamic addresses,
# which we don't really recommend.
#
#subnet 10.254.239.32 netmask 255.255.255.224 {
# range dynamic-bootp 10.254.239.40 10.254.239.60;
# option broadcast-address 10.254.239.31;
# option routers rtr-239-32-1.example.org;
#}

#_A slightly different configuration for an internal subnet.
#
#subnet 172.16.10.0 netmask 255.255.255.0 {
# range 172.16.10.2 172.16.10.254;
# option domain-name-servers ns1.internal.example.org;
# option domain-name "internal.example.org";
# option subnet-mask 255.255.255.0;
# option routers 172.16.10.1;
# option broadcast-address 172.16.10.1;
# default-lease-time 600;
# max-lease-time 7200;
#}

# Hosts which require special configuration options can be listed in
# host statements. If no address is specified, the address will be
# allocated dynamically (if possible), but the host-specific information
# will still come from the host declaration.
#
#host passacaglia {
# hardware ethernet 0:0:c0:5d:bd:95;
# filename "vmunix.passacaglia";
# server-name "toccata.example.com";
#}

[Ctrl]H Help [Ctrl]O Write Out [Ctrl]W Where Is [Ctrl]K Cut [Ctrl]T Execute [Ctrl]C Location [Ctrl]U Undo [Ctrl]M Set Mark [Ctrl]I To Bracket [Ctrl]P Previous
[Ctrl]X Exit [Ctrl]R Read File [Ctrl]N Replace [Ctrl]V Paste [Ctrl]J Justify [Ctrl]G Go To Line [Ctrl]E Redo [Ctrl]Y Copy [Ctrl]O Where Was [Ctrl]N Next
```

Depois salve e saia do arquivo.

6º Passo

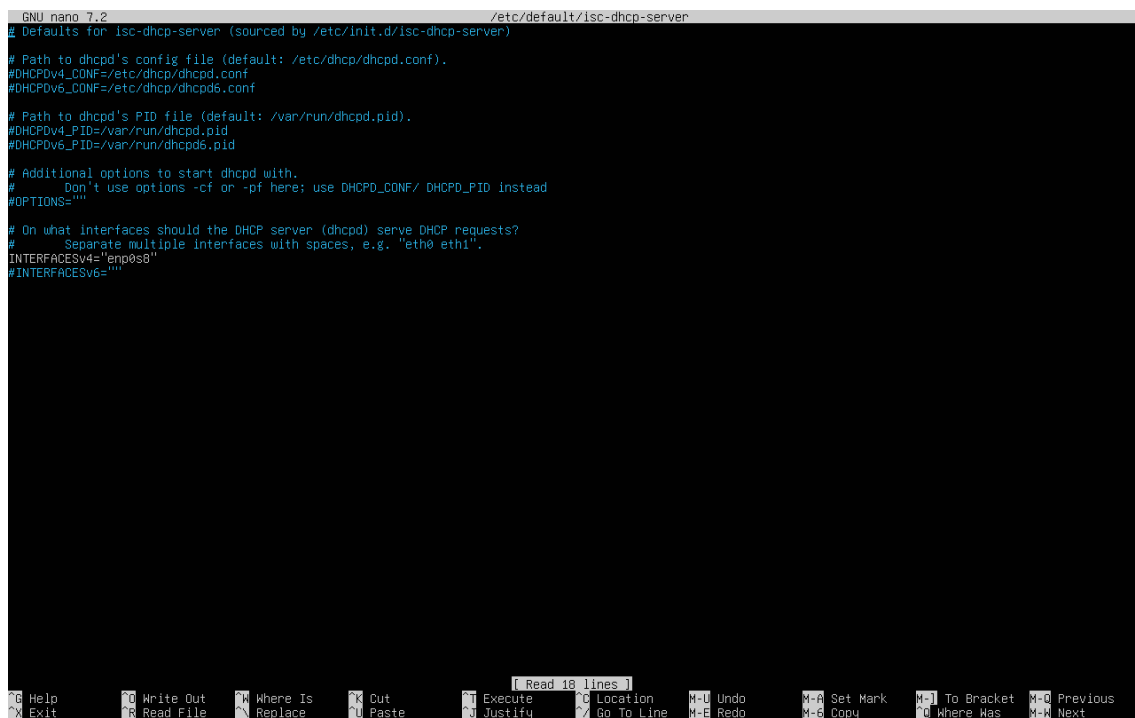
Depois de salvar as configurações, crie uma cópia do “*isc-dhcp-server*” que está no diretório “*default*”:

```
cp /etc/default/isc-dhcp-server /etc/default/isc-dhcp-server.bkp
```

E entre no arquivo original:

```
sudo nano /etc/default/isc-dhcp-server
```

Configure:



```
GNU nano 7.2 /etc/default/isc-dhcp-server
# Defaults for isc-dhcp-server (sourced by /etc/init.d/isc-dhcp-server)

# Path to dhcpd's config file (default: /etc/dhcp/dhcpd.conf).
#DHCPDv4_CONF=/etc/dhcp/dhcpd.conf
#DHCPDv6_CONF=/etc/dhcp/dhcpd6.conf

# Path to dhcpd's PID file (default: /var/run/dhcpd.pid).
#DHCPDv4_PID=/var/run/dhcpd.pid
#DHCPDv6_PID=/var/run/dhcpd6.pid

# Additional options to start dhcpd with.
# Don't use options -cf or -pf here; use DHCPD_CONF/ DHCPD_PID instead
#OPTIONS=""

# On what interfaces should the DHCP server (dhcpd) serve DHCP requests?
# Separate multiple interfaces with spaces, e.g. "eth0 eth1".
INTERFACESv4="enp0s8"
#INTERFACESv6=""

[ Read 18 lines ]
Help      Write Out  Where Is   Cut        Execute    Location   Undo       Set Mark   To Bracket Previous
Exit      Read File  Replace    Paste      Justify    Go To Line Redo      Copy      Where Was  Next
```

A linha “*INTERFACESv6=""*” não vai estar comentada quando acessar a configuração (*isc-dhcp-server*), deixe a comentada, você não irá usar.

7º Passo

Ao terminar de configurar, reinicie o servidor DHCP e verifique seu Status:

sudo systemctl restart isc-dhcp-server

sudo systemctl status isc-dhcp-server

*Em caso de erro, será mostrado uma linha em branco te mostrando a linha que está errada, seja na configuração do seu **isc-dhcp-server**, **netplan** ou **dhcpcd**.*

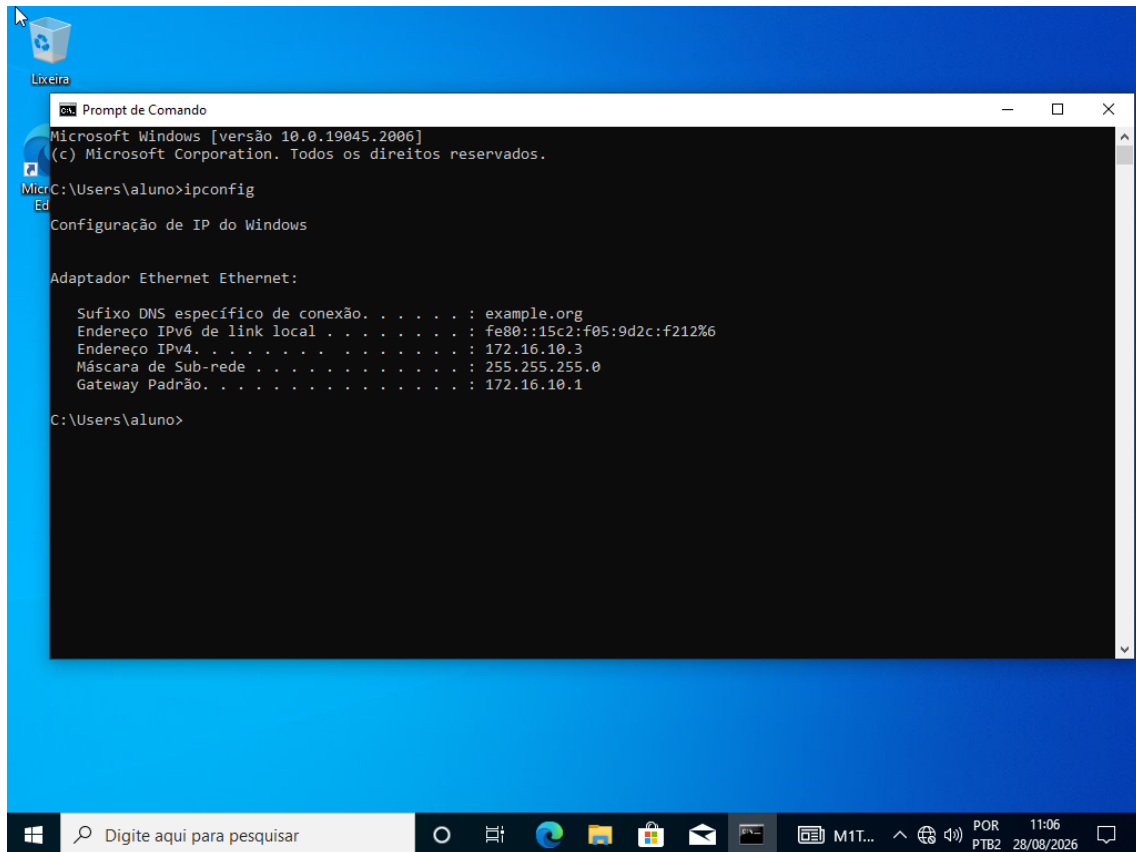
*(Observação: Toda vez que alterar seu arquivo **netplan**, use o **sudo netplan apply**)*

```
aluno@aluno:/home$ sudo systemctl restart isc-dhcp-server
aluno@aluno:/home$ sudo systemctl status isc-dhcp-server
● isc-dhcp-server.service - ISC DHCP IPv4 server
   Loaded: loaded (/usr/lib/systemd/system/isc-dhcp-server.service; enabled; preset: enabled)
   Active: active (running) since Fri 2026-08-28 13:42:03 UTC; 13s ago
     Docs: man:dhcpd(8)
   Main PID: 3181 (dhcpd)
      Tasks: 1 (limit: 1055)
    Memory: 3.8M (peak: 4.2M)
       CPU: 14ms
   CGroup: /system.slice/isc-dhcp-server.service
           └─3181 dhcpd -user dhcpd -group dhcpd -f -4 -pf /run/dhcp-server/dhcpd.pid -cf /etc/dhcp/dhcpd.conf enp0s8

ago 28 13:42:03 aluno dhcpd[3181]: Database file: /var/lib/dhcp/dhcpd.leases
ago 28 13:42:03 aluno dhcpd[3181]: PID file: /run/dhcp-server/dhcpd.pid
ago 28 13:42:03 aluno dhcpd[3181]: Wrote 0 leases to leases file.
ago 28 13:42:03 aluno dhcpd[3181]: Listening on LPF/enp0s8/08:00:27:61:ff:37/172.16.10.0/24
ago 28 13:42:03 aluno sh[3181]: Listening on LPF/enp0s8/08:00:27:61:ff:37/172.16.10.0/24
ago 28 13:42:03 aluno dhcpd[3181]: Sending on LPF/enp0s8/08:00:27:61:ff:37/172.16.10.0/24
ago 28 13:42:03 aluno sh[3181]: Sending on LPF/enp0s8/08:00:27:61:ff:37/172.16.10.0/24
ago 28 13:42:03 aluno dhcpd[3181]: Sending on Socket/fallback/fallback-net
ago 28 13:42:03 aluno sh[3181]: Sending on Socket/fallback/fallback-net
ago 28 13:42:03 aluno dhcpd[3181]: Server starting service.
aluno@aluno:/home$ sudo tail -f /var/log/syslog | grep dhcpd
2026-08-28T13:42:03.480493+00:00 aluno dhcpd[3181]: Database file: /var/lib/dhcp/dhcpd.leases
2026-08-28T13:42:03.480506+00:00 aluno dhcpd[3181]: PID file: /run/dhcp-server/dhcpd.pid
2026-08-28T13:42:03.480524+00:00 aluno dhcpd[3181]: Wrote 0 leases to leases file.
2026-08-28T13:42:03.492602+00:00 aluno dhcpd[3181]: Listening on LPF/enp0s8/08:00:27:61:ff:37/172.16.10.0/24
2026-08-28T13:42:03.492795+00:00 aluno dhcpd[3181]: Sending on LPF/enp0s8/08:00:27:61:ff:37/172.16.10.0/24
2026-08-28T13:42:03.492912+00:00 aluno dhcpd[3181]: Sending on Socket/fallback/fallback-net
2026-08-28T13:42:03.493180+00:00 aluno dhcpd[3181]: Server starting service.
```

8º Passo

Faça o teste em uma outra Máquina Virtual para saber se está funcionando:



```
Prompt de Comando
Microsoft Windows [versão 10.0.19045.2006]
(c) Microsoft Corporation. Todos os direitos reservados.
C:\Users\aluno>ipconfig

Configuração de IP do Windows

Adaptador Ethernet Ethernet:

    Sufixo DNS específico de conexão. . . . . : example.org
    Endereço IPv6 de link local . . . . . : fe80::15c2:f05:9d2c:f212%6
    Endereço IPv4. . . . . : 172.16.10.3
    Máscara de Sub-rede . . . . . : 255.255.255.0
    Gateway Padrão. . . . . : 172.16.10.1

C:\Users\aluno>
```

```

* isc-dhcp-server.service - ISC DHCP IPv4 server
  Loaded: loaded (/usr/lib/systemd/system/isc-dhcp-server.service; enabled; preset: enabled)
  Active: active (running) since Fri 2026-08-28 13:42:03 UTC; 13s ago
    Docs: man:dhcpd(8)
  Main PID: 3181 (dhcpd)
    Tasks: 1 (limit: 1055)
  Memory: 3.8M (peak: 4.2M)
    CPU: 14ms
  CGroup: /system.slice/isc-dhcp-server.service
          └─3181 dhcpd -user dhcpd -group dhcpd -f -4 -pf /run/dhcp-server/dhcpd.pid -cf /etc/dhcp/dhcpd.conf enp0s8

ago 28 13:42:03 aluno dhcpd[3181]: Database file: /var/lib/dhcp/dhcpd.leases
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ago 28 13:42:03 aluno dhcpd[3181]: Wrote 0 leases to leases file.
ago 28 13:42:03 aluno dhcpd[3181]: Listening on LPF/enp0s8/08:00:27:61:ff:37/172.16.10.0/24
ago 28 13:42:03 aluno sh[3181]: Listening on LPF/enp0s8/08:00:27:61:ff:37/172.16.10.0/24
ago 28 13:42:03 aluno dhcpd[3181]: Sending on LPF/enp0s8/08:00:27:61:ff:37/172.16.10.0/24
ago 28 13:42:03 aluno sh[3181]: Sending on LPF/enp0s8/08:00:27:61:ff:37/172.16.10.0/24
ago 28 13:42:03 aluno dhcpd[3181]: Sending on Socket/fallback/fallback-net
ago 28 13:42:03 aluno sh[3181]: Sending on Socket/fallback/fallback-net
ago 28 13:42:03 aluno dhcpd[3181]: Server starting service.
aluno@aluno:~$ sudo tail -f /var/log/syslog | grep dhcpd
2026-08-28T13:42:03.480483+00:00 aluno dhcpd[3181]: Database file: /var/lib/dhcp/dhcpd.leases
2026-08-28T13:42:03.480506+00:00 aluno dhcpd[3181]: PID file: /run/dhcp-server/dhcpd.pid
2026-08-28T13:42:03.480524+00:00 aluno dhcpd[3181]: Wrote 0 leases to leases file.
2026-08-28T13:42:03.492602+00:00 aluno dhcpd[3181]: Listening on LPF/enp0s8/08:00:27:61:ff:37/172.16.10.0/24
2026-08-28T13:42:03.492795+00:00 aluno dhcpd[3181]: Sending on LPF/enp0s8/08:00:27:61:ff:37/172.16.10.0/24
2026-08-28T13:42:03.492912+00:00 aluno dhcpd[3181]: Sending on Socket/fallback/fallback-net
2026-08-28T13:42:03.493180+00:00 aluno dhcpd[3181]: Server starting service.
2026-08-28T13:46:29.197170+00:00 aluno dhcpd[3181]: DHCPDISCOVER from 08:00:27:2e:bf:22 via enp0s8
2026-08-28T13:46:33.160030+00:00 aluno dhcpd[3181]: ns1.example.org: no A record associated with address
2026-08-28T13:46:33.174615+00:00 aluno dhcpd[3181]: ns2.example.org: no A record associated with address
2026-08-28T13:46:33.174718+00:00 aluno dhcpd[3181]: DHCPREQUEST for 172.16.10.2 (172.16.10.0) from 08:00:27:2e:bf:22 (DESKTOP-7HA7570) via enp0s8
2026-08-28T13:46:33.177179+00:00 aluno dhcpd[3181]: DHCPREQUEST for 172.16.10.2 (172.16.10.0) from 08:00:27:2e:bf:22 (DESKTOP-7HA7570) via enp0s8
2026-08-28T13:46:33.239734+00:00 aluno dhcpd[3181]: DHCPACK on 172.16.10.2 to 08:00:27:2e:bf:22 (DESKTOP-7HA7570) via enp0s8
2026-08-28T13:56:38.897567+00:00 aluno dhcpd[3181]: DHCPDISCOVER from 08:00:27:6e:98:66 via enp0s8
2026-08-28T13:56:39.920000+00:00 aluno dhcpd[3181]: ns1.example.org: no A record associated with address
2026-08-28T13:56:39.934638+00:00 aluno dhcpd[3181]: ns2.example.org: no A record associated with address
2026-08-28T13:56:39.934869+00:00 aluno dhcpd[3181]: DHCPREQUEST for 172.16.10.3 to 08:00:27:6e:98:66 (DESKTOP-LEU2I8I) via enp0s8
2026-08-28T13:56:39.936897+00:00 aluno dhcpd[3181]: DHCPREQUEST for 172.16.10.3 (172.16.10.0) from 08:00:27:6e:98:66 (DESKTOP-LEU2I8I) via enp0s8
2026-08-28T13:56:39.959062+00:00 aluno dhcpd[3181]: DHCPACK on 172.16.10.3 to 08:00:27:6e:98:66 (DESKTOP-LEU2I8I) via enp0s8
2026-08-28T14:01:40.533130+00:00 aluno dhcpd[3181]: DHCPREQUEST for 172.16.10.3 from 08:00:27:6e:98:66 (DESKTOP-LEU2I8I) via enp0s8
2026-08-28T14:01:40.559734+00:00 aluno dhcpd[3181]: ns1.example.org: no A record associated with address
2026-08-28T14:01:40.578655+00:00 aluno dhcpd[3181]: ns2.example.org: no A record associated with address
2026-08-28T14:01:40.578797+00:00 aluno dhcpd[3181]: DHCPACK on 172.16.10.3 to 08:00:27:6e:98:66 (DESKTOP-LEU2I8I) via enp0s8
2026-08-28T14:06:39.903070+00:00 aluno dhcpd[3181]: DHCPREQUEST for 172.16.10.3 from 08:00:27:6e:98:66 (DESKTOP-LEU2I8I) via enp0s8
2026-08-28T14:06:39.947343+00:00 aluno dhcpd[3181]: ns1.example.org: no A record associated with address
2026-08-28T14:06:39.958401+00:00 aluno dhcpd[3181]: ns2.example.org: no A record associated with address
2026-08-28T14:06:39.958484+00:00 aluno dhcpd[3181]: DHCPACK on 172.16.10.3 to 08:00:27:6e:98:66 (DESKTOP-LEU2I8I) via enp0s8

```

Servidor funcionando sem problemas!!!